



PROJECT

Course Title : Problem Solving Using Artificial Intelligence (*Penyelesaian Masalah Menggunakan Kecerdasan Buatan*)
Course Code : CSM3601
Instructor : Assoc. Prof. Dr Azreen Azman
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Introduction:

Students are required to develop a complete machine learning solution to solve a real-world problem using supervised or unsupervised learning techniques. The project must include:

- Problem identification
- Data collection or dataset selection
- Exploratory Data Analysis (EDA)
- Data preprocessing
- Feature engineering
- Model development
- Model evaluation
- Result interpretation
- Final presentation and report

Students MUST work in groups of 2–4 students.

Learning Outcomes:

By completing this project, students should be able to:

1. Apply the machine learning workflow to real-world problems.
2. Perform data preprocessing and feature engineering.
3. Compare multiple machine learning algorithms.
4. Evaluate models using suitable metrics.
5. Interpret and communicate machine learning results.
6. Demonstrate critical thinking in model selection and optimization.

Instructions to Students:

1. Install and configure *python* on your machine (See Appendix)
2. Complete all tasks in sequence.
3. Run each code block and observe the output.
4. Answer all exercises in the spaces provided.
5. Save your work as instructed by the lecturer.
6. Submit source code files and screenshots before the end of the session.

Project Requirement:

Phase	Deliverable(s)	Deadline
1: Proposal (10%)	Students must submit: • Project title • Problem statement • Objectives • Proposed dataset • Proposed algorithms • Expected outcome	W8 [21 May]
2: Data Preparation & EDA (20%)	Tasks include: • Data cleaning • Handling missing values • Outlier detection • Visualization • Correlation analysis	W9 – W10 [7 Jun]

	• Feature selection/engineering	
3: Model Development (30%)	Students must: • Train at least 3 machine learning models • Compare performance • Tune hyperparameters • Prevent overfitting Minimum Requirements • Train/test split • Cross-validation • Pipeline implementation	W11-W12 [21 Jun]
4: Evaluation & Discussion (20%)	Suggested Metrics Classification • Accuracy • Precision • Recall • F1-score • ROC-AUC Regression • MAE • MSE • RMSE • R² Score Students should discuss: • Best-performing model • Strengths and weaknesses • Feature importance • Ethical considerations	W13 [28 Jun]
5: Presentation & Demonstration (20%)	Students present: • Problem background • Methodology • Results • Live demo • Challenges faced • Future improvements Deliverables: 1. Slide presentation 2. Short video / animation (max 3 minutes) 3. Poster (size A1) 4. Python codes / Jupyter Notebook	W14 [29 Jun – 3 Jul]

Suggestion for Project Themes:

You may choose any of the following topics or suggest your own ML topic.

1. **Customer Churn Prediction**
Predict whether a customer will leave a service.
2. **Fake News Detection**
Classify news articles as real or fake using NLP techniques.
3. **Heart Disease Prediction**
Predict the likelihood of heart disease using patient medical data.
4. **Retail Sales Forecasting**
Predict future sales using historical sales data.
5. **House Price Prediction**
Predict house prices using property features.
6. **Student Performance Prediction**
Predict student grades based on attendance, study time, internet usage, and participation.
7. **Salary Prediction System**

8. Energy Consumption Prediction
9. Crop Yield Prediction
10. Traffic Flow Prediction
11. Loan Approval Prediction
12. Credit Card Fraud Detection
13. Customer Segmentation for Retail Business
14. Student Learning Behavior Grouping
15. Movie Recommendation Grouping

Recommended Tools

Programming

- Python

Libraries

- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn
- TensorFlow/PyTorch (optional)

Development Environment

- Jupyter Notebook
- Google Colab

Suggested Dataset

- [Kaggle](#)
- [UCI Machine Learning Repository](#)
- [Google Dataset Search](#)
- [Data.gov](#)

Marking Scheme:

Component	Marks
Problem Definition	10
EDA & Data Preparation	20
Model Development	25
Evaluation & Analysis	20
Innovation/Critical Thinking	10
Presentation & Report	15
Total	100

Bonus (10 marks):

Bonus marks will be given for the following:

Students may additionally:

- Build a dashboard
- Deploy using [Streamlit](#)
- Create REST APIs using [FastAPI](#)
- Use cloud deployment
- Integrate generative AI explanations